

Q3)(a) $\because \mathcal{J}$ is finite subset of its 'convex' hull S

$$\Rightarrow \text{given } \sum_{i=1}^M \lambda_i s_i = n \quad \text{where } \sum \lambda_i = 1$$

$$\hat{J}_k(n) = \hat{J}_k \left(\sum_{i=1}^M \lambda_i s_i \right)$$

using convexity, Jensen's inequality

$$\hat{J}_k(n) = \hat{J}_k \left(\sum_{i=1}^M \lambda_i s_i \right) \leq \sum_{i=1}^M \lambda_i \hat{J}_k(s_i)$$

$\because$ this is true for every feasible λ
that includes minimizing one

$$\Rightarrow \hat{J}_k(n) \leq \min \left\{ \sum_{i=1}^M \lambda_i \hat{J}_k(s_i) \mid \lambda_i \in \text{feasible set} \right\}$$

$$\Rightarrow \boxed{\hat{J}_k(n) \leq \tilde{J}_k(n)}$$

(b) using eq (2)

$$J_{N-1}(n) = E \left(g_{N-1}(n, u_{N-1}(n), w_{N-1}) + J_N(f_{N-1}(n, u_{N-1}, w_{N-1})) \right)$$

where $u_{N-1}(n) = \vec{u}$ is policy optimizer

$$\text{given } J_N(f_{N-1}(\cdot)) = \hat{J}_N(\cdot) = \tilde{J}_N(\cdot)$$

plugging this in

$$J_{N-1}(n) = E \left(g_{N-1}(n, u_{N-1}(n), w_{N-1}) + \tilde{J}_N^{v_{N-1}}(f_{N-1}(n, u_{N-1}, w_{N-1})) \right)$$

$u_{N-1} = \vec{u}$ policy optimizing seq

$$J_{N-1}(n) = \min_u E \left(g_{N-1}(n, u, w_{N-1}) + \tilde{J}_N^{v_{N-1}}(f_{N-1}(n, u, w_{N-1})) \right)$$

this is how $\hat{J}_k$ is defined

$$\text{so } J_{N-1}(n) = \hat{J}_{N-1}(n)$$

Using Convexity

$$\hat{J}_{N-1}(n) \leq \tilde{J}_{N-1}^{v_{N-1}}(n) \quad (\text{proved in (d)})$$

$$\text{so } J_{N-1}(n) = \hat{J}_{N-1}(n) \leq \tilde{J}_{N-1}^{v_{N-1}}(n)$$

$$<) \quad J_{N-2}(n) = E \left(g_{N-2}(n, \vec{u}_{N-2}(n), w_{N-2}) + \tilde{J}_{N-1}^{v_{N-1}}(f_{N-1}(n, \vec{u}_{N-2}(n), w_{N-2})) \right)$$

$$\leq E \left(g_{N-2}(n, \vec{u}_{N-2}, w_{N-2}) + \tilde{J}_{N-1}^{v_{N-1}}(f_{N-1}(n, \vec{u}_{N-2}, w_{N-2})) \right)$$

(from part b)

$\therefore \vec{u}_{N-2}$ is optimizing seq.

$$\leq E \left(g_{N-2}(\cdot) + \tilde{J}_{N-1}^{v_{N-1}}(f_{N-1}(\cdot)) \right) = \min_u E \left(g_{N-2}(n, u, w_{N-2}) + \tilde{J}_{N-1}^{v_{N-1}}(f_{N-1}(n, u, w_{N-2})) \right)$$

↓
definition of $\hat{J}_{N-2}$

& convexity implies

$$\hat{J}_{N-2}^{\wedge}(m) \subseteq \hat{J}_{N-2}^{\vee}(m)$$

$$\Rightarrow J_{N-2}(m) \subseteq \hat{J}_{N-2}^{\wedge}(m) \subseteq \hat{J}_{N-2}^{\vee}(m)$$

(d) Proof by induction like we did in lecture 5

Base case $k=N$ ('we go backwards')

$$\text{given } J_N(m) = \hat{J}_N^{\vee}(m) = \hat{J}_N^{\wedge}(m)$$

$$\Rightarrow \text{which is same as } J_N(m) \subseteq \hat{J}_N^{\vee}(m) \subseteq \hat{J}_N^{\wedge}(m) \text{ so true } \checkmark$$

let's assume its true for $k+1$

$$\Rightarrow J_{k+1}(m) \subseteq \hat{J}_{k+1}^{\wedge}(m) \subseteq \hat{J}_{k+1}^{\vee}(m) \quad - (2)$$

using definition

$$\begin{aligned} J_k(m) &= E(g_k(m_k, \vec{u}_k, w_k) + J_{k+1}(f(m_k, \vec{u}_k, w_k))) \\ &\leq E(g_k(m_k, \vec{u}_k, w_k) + \hat{J}_{k+1}^{\vee}(f(m_k, \vec{u}_k, w_k))) \quad (\text{using (2)}) \end{aligned}$$

$$\leq \min_{u \in U_k} E(g_k(m_k, u, w_k) + \hat{J}_{k+1}^{\vee}(f(m_k, u, w_k)))$$

$$\leq \hat{J}_k(m_k) \quad (\text{using definition of } \hat{J})$$

also using conveniently

$$\hat{J}_k(m_k) \subset \tilde{J}_k(m_k)$$

So by induction

$\therefore$ Base case is trivially True
 $\& \quad k^{\text{th}}$ case is true iff $k+1^{\text{th}}$ case is True

We can say that

$$J_k(m) \leq \hat{J}_k(m) \leq \tilde{J}_k(m) \quad \forall m \in S \& \\ \forall k \in \{0, 1, 2, \dots, N\}$$